MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)
THIRD SEMESTER EXAMINATION
(2024 Admission Onwards)

MG3DSCCSC201 – DATA STRUCTURES USING C

Duration: 1.5 hours Maximum Marks: 50

Students should attempt at least one question from each Course Outcome (CO) to enhance their overall outcome attainability.

Part A

Very Short Answer Questions
(Answer **all** questions, each question carries **2** marks.)

1. Identify two examples of linear non-primitive data structures. [U] [CO1]
2. Differentiate between LIFO and FIFO principle. [U] [CO2]
3. Define a binary tree. [K] [CO3]
4. Classify the following as primitive or non-primitive: (a) integer, (b) tree, (c) queue. [U] [CO1]
5. Write the preorder traversal for the binary tree with root A and children B and C. [A] [CO3]

[5 × 2 = 10]

Part B

Short Answer Questions
(Answer **5** out of **7** questions, each question carries **5** marks.)

6. Explain merge sort with an example on the array [6, 3, 8, 2] [U] [CO1]
7. Solve the postfix expression using a stack P=5, 6, 2, +, *, 12, 4, /, - [A] [CO2]
8. Define linked list. Explain its components with a neat diagram. [U] [CO3]
9. State the difference between function definition and function call. [K] [CO1]
10. Explain the terms stack overflow and stack underflow in the context of array-based stack implementations. [U] [CO2]
11. Write the algorithm for traversing a singly linked list and explain with an example. [A] [CO3]
12. Demonstrate the working of merge sort using a small array. [U] [CO1]

[5 × 5 = 25]

Part C

Essay Questions
(Answer 1 out of 2 questions, each question carries 15 marks.)

13. Given an array: A = [12, 23, 35, 45, 64, 87, 90] Search for the element 45 using Binary search. Show the step-by-step process. Also show what happens when the element is not found. [U] [CO1]
14. Construct a binary search tree using the sequence: 45, 25, 60, 15, 30, 50, 70. Show step-by-step insertion and represent the final tree. Perform inorder traversal on it. [A] [CO3]

[1 × 15 = 15]